First-order gain model: observability matrix

1 Model

State equation

$$\delta x_1[k+1] = \delta x_2[k]$$
$$\delta x_2[k+1] = \delta x_3[k]$$
$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k]$$
$$a_x[k+1] = a_x[k]$$

$$F_{dx}[k] = f_{dx}[k] + (1-\lambda_c)\sum_{i=1}^{d} f_{dx}[k-i] \;\approx\; \left(1 + (1-\lambda_c)d\right)\frac{\Delta x_d[k]}{\hat{a}_x[k]} \qquad \text{(constant velocity)}$$

Output equation

$$y_1[k] = \delta x_1[k] + n_x[k], \qquad R_1 = \mathrm{Var}(n_x)$$
$$y_2[k] = a_x[k] + n_a[k], \qquad R_2 = \mathrm{Var}(n_a)$$

Estimation error and output error

$$\mathbf{e}[k] = \begin{bmatrix} e_{\delta x_1}[k] \\ e_{\delta x_2}[k] \\ e_{\delta x_3}[k] \\ e_{a_x}[k] \end{bmatrix} = \begin{bmatrix} \delta x_1[k] - \delta\hat{x}_1[k] \\ \delta x_2[k] - \delta\hat{x}_2[k] \\ \delta x_3[k] - \delta\hat{x}_3[k] \\ a_x[k] - \hat{a}_x[k] \end{bmatrix}, \qquad \begin{aligned} e_{y_1}[k] &= y_1[k] - \delta\hat{x}_1[k] = e_{\delta x_1}[k] + n_x[k] \\ e_{y_2}[k] &= y_2[k] - \hat{a}_x[k] = e_{a_x}[k] + n_a[k] \end{aligned}$$

Error propagation (no noise)

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k], \quad \mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & 1 \end{bmatrix}; \qquad \begin{bmatrix} e_{y_1}[k] \\ e_{y_2}[k] \end{bmatrix} = \mathbf{H}\,\mathbf{e}[k], \quad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2 Observability matrix

Errors at step k_0, output errors j steps later

$$\begin{bmatrix} e_{y_1}[k_0+j] \\ e_{y_2}[k_0+j] \end{bmatrix} = \mathbf{H}\,\mathbf{F}_e[k_0+j-1]\cdots\mathbf{F}_e[k_0]\;\mathbf{e}[k_0]$$

The matrix

$$\begin{bmatrix} e_{y_1}[k_0] \\ e_{y_2}[k_0] \\ e_{y_1}[k_0+1] \\ e_{y_2}[k_0+1] \\ e_{y_1}[k_0+2] \\ e_{y_2}[k_0+2] \\ e_{y_1}[k_0+3] \\ e_{y_2}[k_0+3] \\ e_{y_1}[k_0+4] \\ e_{y_2}[k_0+4] \\ \vdots \\ e_{y_1}[k_0+j] \\ e_{y_2}[k_0+j] \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \lambda_c & -F_{dx}[k_0] \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \lambda_c^2 & -\lambda_c F_{dx}[k_0] - F_{dx}[k_0+1] \\ 0 & 0 & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \lambda_c^{j-2} & -\sum_{i=0}^{j-3} \lambda_c^{j-3-i} F_{dx}[k_0+i] \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\mathbf{O}(k_0)} \begin{bmatrix} e_{\delta x_1}[k_0] \\ e_{\delta x_2}[k_0] \\ e_{\delta x_3}[k_0] \\ e_{a_x}[k_0] \end{bmatrix}$$

The four columns (F_{dx} constant)

$e_{\delta x_1}$: $y_1 : 1, 0, 0, 0, \ldots$ $y_2 : 0, 0, 0, \ldots$

$e_{\delta x_2}$: $y_1 : 0, 1, 0, 0, \ldots$ $y_2 : 0, 0, 0, \ldots$

$e_{\delta x_3}$: $y_1 : 0, 0, 1, \lambda_c, \lambda_c^2, \ldots, \lambda_c^{j-2} \to 0$ $y_2 : 0, 0, 0, \ldots$

e_{a_x} : $y_1 : 0, 0, 0, -F_{dx}, -F_{dx}(1+\lambda_c), \cdots \to -\dfrac{F_{dx}}{1-\lambda_c}$ $y_2 : 1, 1, 1, \ldots$

3 What the matrix shows

y_2 sees only the gain error, the same at every step

$e_{y_2}[k_0+j] = 1 \cdot e_{a_x}[k_0]$ for every j

y_1 sees the gain error only through F_{dx}

$$e_{y_1}[k_0+j]\Big|_{e_{a_x}} \to -\frac{F_{dx}}{1-\lambda_c}\, e_{a_x}[k_0] = -\frac{1+(1-\lambda_c)d}{1-\lambda_c} \cdot \frac{\Delta x_d}{\hat{a}_x}\, e_{a_x}[k_0]$$

hold: $\Delta x_d = 0 \Rightarrow e_{y_1}\Big|_{e_{a_x}} = 0$; faster ($\Delta x_d \uparrow$) or nearer the wall ($\hat{a}_x \downarrow$) $\Rightarrow \left| e_{y_1}\big|_{e_{a_x}} \right| \uparrow$

Where T_d enters

R = particle radius, T_s = sampling period, T_d = descent time

h_0, h_b = start and bottom height of the descent, in units of R

$$\Delta x_d = \frac{\text{distance}}{\text{number of steps}} = \frac{(h_0 - h_b)\, R}{T_d/T_s}$$

y_1 entry of $e_{a_x} \propto \dfrac{1}{T_d\, \hat{a}_x}$, y_2 entry of $e_{a_x} = 1$

4 Solving the matrix equation

4.1 The y_2 rows written out ($\hat{a}_x$ stays at $\hat{a}_x[k_0]$ because $\hat{a}_x[k+1] = \hat{a}_x[k]$)

$$
\begin{aligned}
e_{y_2}[k_0+1] &= y_2[k_0+1] - \hat{a}_x[k_0] = e_{a_x}[k_0] + n_a[k_0+1] \\
e_{y_2}[k_0+2] &= y_2[k_0+2] - \hat{a}_x[k_0] = e_{a_x}[k_0] + n_a[k_0+2] \\
&\vdots \\
e_{y_2}[k_0+N] &= y_2[k_0+N] - \hat{a}_x[k_0] = e_{a_x}[k_0] + n_a[k_0+N]
\end{aligned}
$$

4.2 One number e that fits all N equations best

$$
\begin{aligned}
\text{minimize} \quad & \sum_{j=1}^{N} \big(e_{y_2}[k_0+j] - e\big)^2 \\
\frac{d}{de}: \quad & -2\sum_{j=1}^{N} \big(e_{y_2}[k_0+j] - e\big) = 0 \\
& \hat{e}_{a_x}[k_0] = \frac{1}{N}\sum_{j=1}^{N} e_{y_2}[k_0+j]
\end{aligned}
$$

4.3 Back to the gain

$$
\begin{aligned}
\hat{a}_x[k_0] + \hat{e}_{a_x}[k_0] &= \hat{a}_x[k_0] + \frac{1}{N}\sum_{j=1}^{N} \big(y_2[k_0+j] - \hat{a}_x[k_0]\big) \\
&= \frac{1}{N}\sum_{j=1}^{N} y_2[k_0+j] \\
&= \frac{1}{N}\sum_{j=1}^{N} a_x[k_0+j] + \frac{1}{N}\sum_{j=1}^{N} n_a[k_0+j] \qquad \big(y_2[k_0+j] = a_x[k_0+j] + n_a[k_0+j]\big)
\end{aligned}
$$

4.4 A y_1 row turned into a reading of e_{a_x} (position errors known)

$$
\begin{aligned}
e_{y_1}[k_0+j] &\approx -\frac{F_{dx}}{1-\lambda_c}\, e_{a_x}[k_0] + n_x[k_0+j] \\
-\frac{1-\lambda_c}{F_{dx}}\, e_{y_1}[k_0+j] &\approx e_{a_x}[k_0] - \frac{1-\lambda_c}{F_{dx}}\, n_x[k_0+j], \qquad \text{Var} = \left(\frac{1-\lambda_c}{F_{dx}}\right)^2 R_1
\end{aligned}
$$

4.5 Readings with different noise: weight = 1/Var

$$
\begin{aligned}
y_2 \text{ row:} \quad & \frac{1}{R_2} \\
y_1 \text{ row:} \quad & \frac{1}{R_1}\left(\frac{F_{dx}}{1-\lambda_c}\right)^2 = \frac{1}{R_1}\left(\frac{1+(1-\lambda_c)d}{1-\lambda_c} \cdot \frac{\Delta x_d}{\hat{a}_x}\right)^2
\end{aligned}
$$

4.6 The gain moves, both rows: weighted average

$$\hat{a}_x[k_0] + \hat{e}_{a_x}[k_0] \approx \frac{\sum_{j=1}^{N} w_j \, a_x[k_0 + j]}{\sum_{j=1}^{N} w_j}, \qquad w_j = \frac{1}{R_2} + \frac{1}{R_1}\Big(\frac{F_{dx}[k_0 + j]}{1 - \lambda_c}\Big)^2$$

4.7 What the figure computes (weights read from the filter, initial estimate included)

$$\hat{a}_x[k] = \frac{\dfrac{1}{P_{aa}[0]}\,\hat{a}_x[0] + \sum_{j=1}^{k} w_j \, a_x[j]}{\dfrac{1}{P_{aa}[0]} + \sum_{j=1}^{k} w_j}$$

$$w_j = \frac{1}{P_{aa}[j]} - \frac{1}{P_{aa}[j-1]} \;\approx\; \frac{1}{R_2} + \frac{1}{R_1}\Big(\frac{F_{dx}[j]}{1 - \lambda_c}\Big)^2$$

P_{aa} = variance of the gain estimate

$P_{aa}[0]$ = its initial value

5 Kalman filter: where 4.7 comes from

5.1 Kalman filter, every step

$$\hat{\mathbf{x}}^f[k] = \text{state equation applied to } \hat{\mathbf{x}}[k-1]$$

$$\mathbf{P}^f[k] = \mathbf{F}_e[k-1]\,\mathbf{P}[k-1]\,\mathbf{F}_e^T[k-1] + \mathbf{Q}$$

$$\mathbf{L}[k] = \mathbf{P}^f[k]\,\mathbf{H}^T\big(\mathbf{H}\,\mathbf{P}^f[k]\,\mathbf{H}^T + \mathbf{R}\big)^{-1}$$

$$\hat{\mathbf{x}}[k] = \hat{\mathbf{x}}^f[k] + \mathbf{L}[k]\big(\mathbf{y}[k] - \hat{\mathbf{y}}[k]\big), \qquad \hat{\mathbf{y}}[k] = \mathbf{H}\,\hat{\mathbf{x}}^f[k]$$

$$\mathbf{P}[k] = \big[\mathbf{I} - \mathbf{L}[k]\,\mathbf{H}\big]\,\mathbf{P}^f[k]$$

$\mathbf{P}$ = error covariance, $\quad \mathbf{Q} = \text{diag}(0,\, 0,\, Q_{33},\, 0), \quad \mathbf{R} = \text{diag}(R_1,\, R_2)$

5.2 The a_x element, y_2 only

$$\hat{a}_x^f[k] = \hat{a}_x[k-1]$$

$$P_{aa}^f[k] = P_{aa}[k-1] + Q_{44} = P_{aa}[k-1]$$

$$\ell_{42}[k] = \frac{P_{aa}^f[k]}{P_{aa}^f[k] + R_2}$$

$$\hat{a}_x[k] = \hat{a}_x^f[k] + \ell_{42}[k]\,\big(y_2[k] - \hat{a}_x^f[k]\big)$$

$$P_{aa}[k] = \big(1 - \ell_{42}[k]\big)\,P_{aa}^f[k]$$

5.3 P_{aa} written as $1/P_{aa}$

$$P_{aa}[k] = \frac{R_2}{P_{aa}[k-1] + R_2}\,P_{aa}[k-1]$$

$$\frac{1}{P_{aa}[k]} = \frac{P_{aa}[k-1] + R_2}{R_2\,P_{aa}[k-1]} = \frac{1}{P_{aa}[k-1]} + \frac{1}{R_2}$$

5.4 $\hat{a}_x$ written as $\hat{a}_x/P_{aa}$

$$\ell_{42}[k] = \frac{P_{aa}[k-1]}{P_{aa}[k-1] + R_2} = \frac{P_{aa}[k]}{R_2}, \qquad 1 - \ell_{42}[k] = \frac{R_2}{P_{aa}[k-1] + R_2} = \frac{P_{aa}[k]}{P_{aa}[k-1]}$$

$$\hat{a}_x[k] = \big(1 - \ell_{42}[k]\big)\,\hat{a}_x[k-1] + \ell_{42}[k]\,y_2[k] = P_{aa}[k]\left(\frac{\hat{a}_x[k-1]}{P_{aa}[k-1]} + \frac{y_2[k]}{R_2}\right)$$

$$\frac{\hat{a}_x[k]}{P_{aa}[k]} = \frac{\hat{a}_x[k-1]}{P_{aa}[k-1]} + \frac{y_2[k]}{R_2}$$

5.5 Unroll from step 0 to step k

$$\frac{1}{P_{aa}[k]} = \frac{1}{P_{aa}[0]} + \sum_{j=1}^{k} \frac{1}{R_2}$$

$$\frac{\hat{a}_x[k]}{P_{aa}[k]} = \frac{\hat{a}_x[0]}{P_{aa}[0]} + \sum_{j=1}^{k} \frac{1}{R_2}\, y_2[j]$$

$$\hat{a}_x[k] = \frac{\dfrac{1}{P_{aa}[0]}\,\hat{a}_x[0] + \sum_{j=1}^{k} \dfrac{1}{R_2}\, y_2[j]}{\dfrac{1}{P_{aa}[0]} + \sum_{j=1}^{k} \dfrac{1}{R_2}}$$

$$= 4.7 \text{ with } w_j = \frac{1}{R_2} = \frac{1}{P_{aa}[j]} - \frac{1}{P_{aa}[j-1]}, \qquad y_2[j] = a_x[j] + n_a[j]$$

5.6 y_1 added: the e_{a_x} column of O is carried in $\mathbf{P}^f$

$$P^f_{3a}[k] = \lambda_c\, P_{3a}[k-1] - F_{dx}\, P_{aa}[k-1], \qquad P^f_{2a}[k] = P_{3a}[k-1], \qquad P^f_{1a}[k] = P_{2a}[k-1]$$

$$P^f_{1a} \to -\frac{F_{dx}}{1-\lambda_c}\, P_{aa} \qquad \text{(the } y_1 \text{ entry of the } e_{a_x} \text{ column, times } P_{aa})$$

$$\ell_{41}[k] = \frac{P^f_{a1}[k]}{P^f_{11}[k] + R_1}, \qquad \frac{1}{P_{aa}[k]} - \frac{1}{P_{aa}[k-1]} \approx \frac{1}{R_2} + \frac{1}{P^f_{11}[k] + R_1} \left(\frac{F_{dx}}{1-\lambda_c}\right)^2 = w_k$$

6 One step, two tracks; the estimate unrolled

6.1 Track 1: confidence and gain (no measurement value enters)

$$P_{aa}^{f}[k] = P_{aa}[k-1] + Q_{44}$$

$$\ell_{42}[k] = \frac{P_{aa}^{f}[k]}{P_{aa}^{f}[k] + R_2}$$

$$P_{aa}[k] = \big(1 - \ell_{42}[k]\big)\, P_{aa}^{f}[k]$$

6.2 Track 2: estimate (the measurement enters here only)

$$\hat{a}_x^{f}[k] = \hat{a}_x[k-1]$$

$$e_{y_2}[k] = y_2[k] - \hat{a}_x^{f}[k]$$

$$\hat{a}_x[k] = \hat{a}_x^{f}[k] + \ell_{42}[k]\, e_{y_2}[k] = \big(1 - \ell_{42}[k]\big)\, \hat{a}_x[k-1] + \ell_{42}[k]\, y_2[k]$$

6.3 Unroll track 2

$$\hat{a}_x[1] = (1 - \ell_{42}[1])\, \hat{a}_x[0] + \ell_{42}[1]\, y_2[1]$$

$$\hat{a}_x[2] = (1 - \ell_{42}[2])(1 - \ell_{42}[1])\, \hat{a}_x[0] + (1 - \ell_{42}[2])\, \ell_{42}[1]\, y_2[1] + \ell_{42}[2]\, y_2[2]$$

$$\begin{aligned}\hat{a}_x[3] = {} & (1 - \ell_{42}[3])(1 - \ell_{42}[2])(1 - \ell_{42}[1])\, \hat{a}_x[0] + (1 - \ell_{42}[3])(1 - \ell_{42}[2])\, \ell_{42}[1]\, y_2[1] \\ & + (1 - \ell_{42}[3])\, \ell_{42}[2]\, y_2[2] + \ell_{42}[3]\, y_2[3]\end{aligned}$$

$$\hat{a}_x[k] = c_{k,0}\, \hat{a}_x[0] + \sum_{j=1}^{k} c_{k,j}\, y_2[j], \qquad c_{k,j} = \ell_{42}[j] \prod_{i=j+1}^{k} \big(1 - \ell_{42}[i]\big), \qquad c_{k,0} + \sum_{j=1}^{k} c_{k,j} = 1$$

6.4 $Q_{44} = 0$: every coefficient equal (average)

$$\ell_{42}[i] = \frac{1}{i+m}, \quad m = \frac{R_2}{P_{aa}[0]}, \qquad 1 - \ell_{42}[i] = \frac{i+m-1}{i+m}$$

$$\prod_{i=j+1}^{k} \big(1 - \ell_{42}[i]\big) = \frac{j+m}{j+m+1} \cdot \frac{j+m+1}{j+m+2} \cdots \frac{k+m-1}{k+m} = \frac{j+m}{k+m}$$

$$c_{k,j} = \frac{1}{j+m} \cdot \frac{j+m}{k+m} = \frac{1}{k+m} \qquad \text{(same for every } j\text{)}$$

6.5 $Q_{44} > 0$: coefficients shrink with age (forgetting)

$$\ell_{42}[k] \to \ell^* \approx \sqrt{Q_{44}/R_2} \qquad (Q_{44} \ll R_2)$$

$$c_{k,j} \approx \ell^* \left(1 - \ell^*\right)^{k-j}, \qquad \text{memory} \approx \frac{1}{\ell^*} = \sqrt{R_2/Q_{44}} \text{ steps}$$

Descent only, $T_d = 5$ s vs 50 s: filter $\hat{a}_x$ (blue) vs the boxed formula of 4.7 (green dashed)

